

Tier 1 (<5 LPA)

Q1. Traffic Fine System

Story:

A smart traffic system tracks overspeeding vehicles. If speed > 80 → fine ₹500. If >100 → ₹1000.
If repeated violation → double fine.

Tag: TCS NQT 2023

Input:

Speed = 105

Repeated = Yes

Output:

Fine = 2000

Q2. Smart Electricity Billing

Story:

Electricity bill is calculated:

- Units $\leq 100 \rightarrow ₹5/\text{unit}$
 - 101–200 $\rightarrow ₹7/\text{unit}$
 - 200 $\rightarrow ₹10/\text{unit}$
- If bill > ₹2000 $\rightarrow 10\%$ surcharge

Tag: Infosys 2022

Input:

Units = 250

Output:

Bill = 2750

Surcharge = 275

Total = 3025

Q3. Login Attempt Lock

Story:

User gets max 3 attempts. If correct → "Login Successful". If 3 wrong → "Account Locked".

Tag: *Wipro 2023*

Input:

Attempts: wrong, wrong, wrong

Output:

Account Locked

Q4. Discount Eligibility**Story:**

If amount $\geq 5000 \rightarrow 20\%$ discount

If $\geq 3000 \rightarrow 10\%$

Else $\rightarrow$ no discount

If user is premium $\rightarrow$ extra 5%

Tag: *Capgemini 2022*

Input:

Amount = 6000

Premium = Yes

Output:

Final Amount = 4500

Q5. Odd-Even Game**Story:**

Game checks numbers till 0 is entered. Count odd & even numbers.

Tag: *Cognizant 2023*

Input:

2 3 4 5 0

Output:

Even = 2

Odd = 2

Tier 2 :

Q6. ATM Withdrawal Logic

Story:

Withdraw only if:

- Balance $\geq$ amount
 - Minimum balance ₹1000 maintained
- Else show reason

Tag: *Accenture 2024*

Input:

Balance = 5000

Withdraw = 4500

Output:

Transaction Failed: Minimum balance violation

Q7. Cab Fare System

Story:

Fare:

- First 5 km $\rightarrow$ ₹50/km
 - Next 5 $\rightarrow$ ₹40/km
 - Beyond $\rightarrow$ ₹30/km
- Night $\rightarrow$ +20%

Tag: *Uber 2023*

Input:

Distance = 12

Night = Yes

Output:

Fare = 624

Q8. Student Result Analyzer**Story:**

Fail if any subject <33

Else:

- Avg $\geq 75 \rightarrow$ Distinction
- $\geq 60 \rightarrow$ First Class
- $\geq 50 \rightarrow$ Second
Else Pass

Tag: *HCL 2023*

Input:

Marks: 80 70 60 90 85

Output:

Distinction

Q9. Password Strength Checker**Story:**

Password must have:

- ≥ 8 chars
 - 1 digit
 - 1 uppercase
 - 1 special char
- Else show missing criteria

Tag: *Tech Mahindra 2024*

Input:

Password = Abc123

Output:

Weak Password: Missing special character, length < 8

Q10. Train Ticket Booking

Story:

Seats available → confirm

Else → waitlist

If VIP → override

Tag: *IRCTC (Practice Scenario) 2023*

Input:

Seats = 0

VIP = Yes

Output:

Ticket Confirmed



Tier 3 (>9 LPA)

Q11. Fraud Transaction Detection

Story:

Transaction flagged if:

- Amount > ₹50,000 AND location mismatch
- OR >3 transactions in 1 min

Tag: *Amazon 2024*

Input:

Amount = 60000

LocationMatch = No

Transactions = 1

Output:

Fraud Detected

Q12. Dynamic Pricing Engine

Story:

Price increases:

- Demand High → +20%
- Weekend → +10%
- Both → cumulative

Tag: *Flipkart 2024*

Input:

Base = 1000

Demand = High

Weekend = Yes

Output:

Final Price = 1320

Q13. Smart Elevator System

Story:

Elevator moves:

- Same direction requests first
- Then opposite
- Skip overloaded condition

Tag: *Microsoft 2023*

Input:

Current Floor = 5

Requests: 7(up), 3(down)

Output:

Next Stop = 7

Q14. E-commerce Return Policy

Story:

Return allowed if:

- ≤ 7 days
- OR defective
- Premium users $\rightarrow$ 15 days

Tag: *Amazon 2023*

Input:

Days = 10

Defective = No

Premium = Yes

Output:

Return Accepted

Q15. Online Exam Anti-Cheat System

Story:

Flag student if:

- Tab switch > 3 times
- OR no activity for > 5 min

Tag: *Google 2024*

Input:

TabSwitch = 4

IdleTime = 2

Output:

Cheating Suspected
